

Game Theory – 1

Impartial games: P-positions and the Sprague–Grundy theory

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Theoretical minimum. A complete theory exists for one class of games only, and it is worth knowing exactly where that class ends. Everything below is proved here.

Definitions. A game is *impartial* if from every position both players have the same set of moves — the moves depend on the position, not on whose turn it is. It is *finite* if the game graph has no infinite path, and we play the *normal* convention: the player unable to move loses (the opposite convention is called *misère*). A position is an *N-position* if the player to move (Next) wins, and a *P-position* if the player who has just moved (Previous) wins.

Theorem 1 (Zermelo; backward induction). In a finite impartial game with no draws exactly one of the two players has a winning strategy, and the positions split uniquely into P and N by the rules: a terminal position is a P-position; a position is an N-position if and only if some move leads to a P-position; a position is a P-position if and only if every move leads to an N-position. *Proof:* induct on the length of the longest play from the position; the three rules determine the label of a position from the labels of its successors, and the strategy “always move to a P-position” wins.

Theorem 2 (Bouton: the theory of Nim). In Nim a move takes any positive number of tokens from one heap. The position $(a_1, \dots, a_k)$ is a P-position if and only if $a_1 \oplus a_2 \oplus \dots \oplus a_k = 0$, where $\oplus$ is bitwise exclusive or (addition in $\mathbb{F}_2^m$ of the binary expansions). *Proof:* two lemmas. If the sum is 0, any move changes exactly one heap and therefore makes the sum non-zero. If the sum s is non-zero, look at its highest non-zero bit; some heap a_i has that bit set, and $a_i \oplus s < a_i$, so one may reduce that heap to $a_i \oplus s$ and obtain sum 0. The terminal position has sum 0, and Theorem 1 applies.

Definition (Grundy value). For a finite set S of non-negative integers let $\text{mex } S$ be the least non-negative integer not in S . The *Grundy value* of a position x is defined by recursion: $g(x) = \text{mex}\{g(y) : x \rightarrow y\}$. A terminal position has $g = 0$, and by Theorem 1 a position is a P-position if and only if $g(x) = 0$. A heap of n tokens in Nim has $g = n$.

Theorem 3 (Sprague–Grundy). The *sum* $G_1 + G_2$ of two games is the game in which a move consists in making a move in exactly one component. Then

$$g(x_1, x_2) = g(x_1) \oplus g(x_2).$$

Proof: by induction on the sum of the lengths. Put $s = g(x_1) \oplus g(x_2)$. First, no move from (x_1, x_2) leads to a position of Grundy value s , exactly as in the first lemma of Theorem 2. Second, for every $t < s$ some move produces Grundy value t : take the highest bit where t and s differ, choose the component whose Grundy value has that bit set, and use the mex property to move there. Hence the mex is s . *Consequence:* every finite impartial game is equivalent to a single Nim heap of g tokens, and a sum of games is won exactly when the $\oplus$ of the Grundy values is non-zero.

Fact 1 (subtraction games and misère play). If the allowed moves are “remove s tokens” for s in a finite set S , the Grundy sequence $g(0), g(1), g(2), \dots$ is eventually periodic, because $g(n)$ depends only on the previous $\max S$ values and there are finitely many such windows. Misère play is genuinely different and has no general theory; for Nim, however, the rule is simple: with all heaps of size 1 the P-positions are those with an even number of heaps, and otherwise the normal rule $\oplus = 0$ still decides.

1. Prove Theorem 1 and the labelling rules. Then compute the P-positions of the subtraction game with $S = \{1, 2, 3\}$ and of Bachet’s game with $S = \{1, \dots, k\}$, and prove that for an arbitrary finite S the sequence of labels is eventually periodic with period at most $2^{\max S}$.
2. Prove Bouton’s theorem in full, find all winning first moves in the Nim

position $(3, 5, 7)$, and prove the misère rule stated in Fact 1.

3. A game starts with four heaps of beans, containing 3, 4, 5 and 6 beans. The two players move alternately. A move consists of taking either (a) one bean from a heap, provided at least two beans are left behind in that heap, or (b) a complete heap of two or three beans. The player who takes the last heap wins. To win the game, do you want to move first or second? Give a winning strategy. (Putnam 1995)
4. Alice and Bob take turns removing stones from a heap that initially has n stones; the number of stones removed at each turn must be one less than a prime number, and the winner is the player who takes the last stone. Alice plays first. Prove that there are infinitely many n for which Bob has a winning strategy. (Putnam 2006)
5. Prove the Sprague–Grundy theorem. Then compute the Grundy values of the subtraction game with $S = \{1, 3, 4\}$, decide who wins the sum of three such games with 10, 12 and 15 tokens, and explain why the theorem says nothing about a game in which the two players have different move sets.